

SMART PARKING AVAILABILITY & PREDICTION SYSTEM

A Computer-Vision-Assisted, ML-Driven Parking Guidance Platform for Megaworld Townships

Consolidated Project Scope & Implementation Plan
Prepared for Internship Proposal Review
Version 1.0 | Draft for Discussion

Target Deployment Context	Megaworld townships (e.g., Uptown BGC, Eastwood City) — integrated into existing township web/mobile app
Core Capability	Real-time, slot-level parking availability + predictive forecasting for future arrival times
Key Technologies	Computer vision (occupancy + ANPR), time-series ML forecasting, ticketing system integration

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1. Project Overview

This project proposes a smart parking availability and prediction system for Megaworld townships, beginning with a pilot deployment in one property (e.g., Uptown BGC or Eastwood City) with an architecture designed to scale across other townships. The system has two core pillars:

- **Real-time slot-level availability** — a “movie-seat-style” interface showing which exact parking slot, on which level, is currently free or occupied, powered by wide-angle camera-based occupancy detection.
- **Predictive forecasting** — a machine learning layer that estimates the probability of parking availability at a future time (e.g., “will there be a slot open by 6:30 PM today?”), accounting for historical patterns, day/time, and known external factors such as events and holidays.

The system additionally integrates with the existing parking ticketing/payment system, using approximate license plate matching to infer when a specific vehicle has settled its parking fee and is likely to vacate its slot soon — providing an early signal ahead of the slot physically becoming free.

The intended integration point is Megaworld's existing township web app, meaning this system is designed as a feature module (API + UI component) rather than a standalone application.

2. Problem Statement

Visitors to dense mixed-use townships such as Uptown BGC and Eastwood City frequently cannot tell, before arriving, whether parking will be available in a given building or level. This is especially problematic:

- During peak hours (lunch, after-office, weekend mall hours)
- During scheduled events, concerts, holidays, or promotional activities
- When visitors are commuting from other townships and must decide, mid-trip, whether to detour to an alternate parking structure

Today, this information is invisible until the driver physically arrives and circles the building — wasting time, fuel, and creating congestion at entry points. There is currently no forward-looking, data-driven way to answer “will I be able to park by the time I get there?”

3. Project Objectives

1. Provide real-time, slot-level parking visibility (occupied/free, by level and zone) through a seat-map-style interface embedded in the existing Megaworld web app.
2. Build a predictive model that forecasts parking availability at a specified future time and location, factoring in historical occupancy trends, time-of-day/day-of-week patterns, holidays, and scheduled events.
3. Automatically infer near-term slot vacancy by cross-referencing camera-detected license plates against the parking ticketing/payment system, without requiring high-precision plate recognition.
4. Design the system as a modular feature that can be integrated into and scaled across multiple Megaworld townships, not just a single pilot site.
5. Keep the computer vision and inference approach lightweight and computationally efficient, avoiding unnecessary accuracy/cost trade-offs (e.g., full plate-perfect OCR) in favor of “good enough, contextually narrowed” matching.

4. Non-Goals / Explicit Scope Boundaries

To keep the internship project scoped realistically, the following are explicitly out of scope for the initial version:

- Slot reservation or booking (the seat-map UI shows availability only — it is informational, not transactional, similar to a live seat map without the ability to reserve a seat)
- Dynamic/surge pricing based on predicted demand
- Full building-wide 100% camera coverage on day one — pilot scope will cover a representative subset of levels/zones
- Automated payment enforcement or violation flagging via CV (this system observes and predicts; it does not police)
- Cross-property routing/navigation (e.g., turn-by-turn directions) — the system informs the decision, not the driving route itself

5. System Architecture Overview

The system is composed of four layers that map cleanly onto the data flow from physical parking lot to the end-user interface:

5.1 Layer Summary

Layer	Responsibility	Primary Output
Sensing Layer	Wide-angle cameras detect per-slot occupancy and read license plates opportunistically	Occupancy state per slot; candidate plate reads
Data / Integration Layer	Cross-references CV output with ticketing/payment records; stores historical occupancy log	Unified occupancy + payment-status dataset
Prediction Layer (ML)	Forecasts future occupancy per zone/level given time, date, and contextual features	Availability probability for a future timestamp
Application / UI Layer	Presents current + predicted availability in the existing Megaworld web app	Seat-map view, trip-planning availability estimate

Design Principle:

Each layer should degrade gracefully. If CV occupancy fails for a zone, the UI should fall back to a coarser row/level-level estimate rather than showing nothing. If the ML model lacks enough data for a new site, it should fall back to a simple historical-average heuristic rather than blocking the feature.

6. Sensing Layer: Computer Vision

6.1 Occupancy Detection

Wide-angle cameras are mounted to cover an aisle or row of slots (not one camera per stall). Each camera's field of view is calibrated once, at setup time, to map fixed regions of the frame to specific slot IDs. The occupancy model only needs to answer a binary question per region — “is a vehicle present in this bounding area or not” — which is a comparatively low-difficulty CV task and tolerant of moderate lighting/angle variation.

- **Coverage unit:** one camera → one row/aisle → multiple slots, calibrated once per camera install.

- **Update frequency:** near-real-time (e.g., every few seconds to ~1 minute), independent of plate-reading success.
- **Resilience principle:** occupancy status must not depend on a successful plate read — it is derived purely from vehicle presence detection.

6.2 License Plate Recognition (ANPR) — “Good Enough,” Not Perfect

Unlike occupancy detection, reading plates accurately at a wide angle is harder and more resource-intensive. Rather than pursuing high-accuracy OCR on every frame, the system treats plate reads as low-confidence signals to be resolved through progressive narrowing against a small, already-known candidate set — not a full database search.

- **Why the candidate pool is small:** the system already knows, from the ticketing system, which plates are currently checked into the structure and unresolved (i.e., paid but not yet confirmed vacated). Matching only needs to search within this small pool, not all plates ever recorded.
- **Progressive narrowing approach:** partial or low-confidence OCR reads (e.g., first 3 characters, or first/last digits) are used to filter the candidate pool step-by-step until a best match emerges or the pool is reduced to a single plausible candidate.
- **Confidence-aware matching:** each OCR read carries a per-character confidence score; matching logic should weigh mismatches on low-confidence characters more leniently than mismatches on high-confidence characters, and treat commonly confused character pairs (e.g., 0/O, 1/I, 8/B, 5/S, 2/Z) as low-cost substitutions rather than full mismatches.
- **Threshold-gated resolution:** a match is only accepted above a minimum confidence threshold. Below threshold, or in a near-tie between two plausible candidates, the system leaves the slot as “occupied, payment unresolved” rather than guessing — since misidentifying a vacancy is more costly than simply not knowing yet.

6.3 Practical Constraints to Plan For

- Occlusion from pillars, adjacent vehicles, or oversized vehicles (vans/SUVs) blocking neighboring slots in the frame
- Lighting variability across levels (basement levels vs. open-air upper levels)
- Camera placement and calibration is a one-time but non-trivial setup task per level — should be scoped explicitly as a deployment step, not assumed to be automatic
- A small, accepted blind-spot/error rate is standard even in commercial camera-based guidance systems — 100% coverage is not the goal

7. Integration Layer: Ticketing & Payment Cross-Reference

The parking ticketing system already records, at minimum, plate number (or ticket ID), entry timestamp, and payment/settlement timestamp. This system is treated as the authoritative source of payment status, while the CV layer supplies real-time physical presence.

7.1 Status Model (Slot-Level)

Rather than a simple binary occupied/free, the slot status model should support an intermediate state that reflects the true sequence of events (payment does not equal physical vacancy):

Status	Meaning	Trigger
Free	No vehicle detected in slot	CV occupancy detection = empty
Occupied — Unpaid	Vehicle present, no matching settled payment found	CV occupancy detection = present; no plate match
Occupied — Likely Vacating Soon	Vehicle present, payment confirmed settled via plate match	CV occupancy + matched, above-threshold plate-to-payment match
Free (Confirmed)	Slot transitions back to free once vehicle physically leaves	CV occupancy detection transitions present → empty

This distinction directly improves the UI's usefulness — a driver can be shown not just “taken,” but “likely opening up soon,” which is genuinely actionable information when trip-planning.

7.2 Data Captured for Downstream Use

- Every occupancy transition (free ↔ occupied) is logged with slot ID, level, zone, and timestamp
- Every payment-to-plate match (successful or unresolved) is logged with confidence score and outcome
- This log becomes the historical dataset used to train and continuously improve the prediction layer (Section 8)

8. Prediction Layer: Machine Learning Forecasting

This is the layer that differentiates the project from a simple live-occupancy display: predicting whether a slot (or set of slots) will likely be available at a future point in time — the moment the user is expected to arrive.

8.1 What the Model Predicts

- **Primary prediction target:** probability of availability (or expected number of free slots) for a given zone/level at a specified future timestamp — e.g., “Eastwood Mall Basement 2, 6:30 PM today.”
- **Secondary/derived insights:** typical peak-hour windows per day-of-week, per site; anomalous demand spikes correlated to detected events or holidays.

8.2 Categories of Input Signal

Category	Examples	Source
Historical occupancy patterns	Hour-of-day, day-of-week, rolling trend for the same zone	Internal occupancy log (Section 7.2)
Calendar context	Public holidays, weekends, long-weekend clustering	Public holiday calendar (static/lookup)
Scheduled events	Concerts, bazaars, mall promos, building-specific events	Megaworld events calendar / township event listings
Weather (optional, later phase)	Rain, which affects walk-in vs. drive-in behavior	Public weather API
Site metadata	Total capacity, zone size, proximity to anchor tenants (e.g., mall entrance, office lobby)	Static site configuration

8.3 Cold-Start Strategy

A new site or a newly instrumented zone will have no historical data on day one. The plan should explicitly account for this:

- **Phase A — Heuristic baseline:** before sufficient historical data exists, the system falls back to simple rule-based estimates (e.g., generic weekday/weekend/lunch-hour occupancy curves, adjustable by an admin).
- **Phase B — Learned model:** once enough historical occupancy data accumulates (a minimum data-collection period should be defined, e.g., 4–8 weeks per site), the system transitions to a trained forecasting model.
- **Phase C — Continuous refinement:** the model is periodically retrained as new data accumulates, and event/holiday effects are refined based on observed (not just assumed) impact.

8.4 Evaluation Approach (Conceptual)

Model quality should be judged less by generic accuracy metrics and more by whether its predictions are actionable and trustworthy for trip-planning:

- How far in advance can the model make a reliable prediction (e.g., 30 minutes vs. 2 hours out)?
- How does prediction accuracy degrade during atypical days (events, holidays) versus typical days?
- Are “available” predictions rarely wrong in the optimistic direction (i.e., the model should be conservative — telling someone a spot will likely be open when it won't is worse than the reverse)?

9. Application Layer: User Interface

9.1 Core UI Concept — “Seat Map, Minus Reservation”

The interface mirrors the familiar mental model of a movie-seat selector: users see a visual layout of the parking structure, organized by level and zone/row, with each slot color-coded by status. Unlike a movie seat map, there is no selection or booking action — it is purely informational.

- **Current view:** live slot-level status (Free / Occupied — Unpaid / Likely Vacating Soon), filterable by level.
- **Planning view:** a simplified, forward-looking estimate — e.g., “Estimated availability if you arrive at 6:30 PM: Likely available” — intended for the trip-planning use case (deciding whether to head to Uptown or Eastwood before leaving).

9.2 Integration Points into Existing Megaworld Web App

- Exposed as a feature module/tab within the existing township app rather than a separate application
- Reuses existing user session/authentication if available, to avoid duplicating login systems
- API-first design so the same availability/prediction data can later power a widget, notification, or map pin, not just the dedicated screen

9.3 Illustrative User Journeys

Journey A — Live arrival: A user already near Eastwood opens the app, sees the live seat map for the mall parking building, and picks a level with visibly free slots.

Journey B — Trip planning: A user in Uptown is deciding whether to drive to Eastwood for dinner. They check the app's predicted availability for Eastwood at their expected arrival time (e.g., 7:00 PM) before leaving, factoring in that a concert is scheduled that evening.

10. Data Requirements & Privacy Considerations

10.1 Data Involved

Data Type	Sensitivity	Notes
Slot occupancy status	Low	No personal data; safe to expose broadly
License plate images/reads	High	Considered personal data under PH Data Privacy Act
Plate-to-payment match records	High	Links a plate to a payment/time; should be access-restricted
Aggregated historical occupancy	Low	Safe for model training and analytics

10.2 Recommended Practices

- Avoid storing raw plate images longer than necessary — retain only the matching result and a short-lived reference, not a permanent image archive

- Restrict access to plate-linked records to systems that need it (payment cross-referencing), not general analytics or the public-facing UI
- Anonymize or aggregate data used for ML training so the model does not need identity-linked information
- Flag this system for review against the Philippines' Data Privacy Act (RA 10173) and Megaworld's internal data governance policy before any real deployment beyond a sandboxed pilot

11. Phased Implementation Roadmap

Given a typical internship timeline, the project is structured into phases that each produce a demonstrable milestone, rather than requiring the full system to work end-to-end before anything can be shown.

Phase 0 — Discovery & Scoping (Weeks 1–2)

- Confirm pilot site and available camera/ticketing infrastructure with Megaworld IT/Operations
- Define data access to existing ticketing system (read access, format, update frequency)
- Define slot/zone/level taxonomy for the pilot site

Phase 1 — Live Occupancy MVP (Weeks 3–6)

- Camera calibration for a representative subset of slots (one or two rows/levels)
- Occupancy detection pipeline producing Free/Occupied status per calibrated slot
- Seat-map UI prototype consuming this live status (can be a standalone prototype before full app integration)

Phase 2 — Ticketing Cross-Reference (Weeks 5–8, overlapping Phase 1)

- Integrate read access to ticketing/payment records
- Implement plate-read-to-payment progressive matching against the current candidate pool
- Add “Likely Vacating Soon” status to the UI

Phase 3 — Historical Data Collection (Ongoing from Phase 1 onward)

- Begin continuous logging of occupancy transitions from the moment Phase 1 is live
- This running window is what will eventually train the prediction model — the earlier logging starts, the sooner Phase 4 becomes viable

Phase 4 — Prediction Layer v1 (Weeks 8–12, contingent on data volume)

- Start with heuristic baseline (historical hourly/daily averages) if data volume is still thin
- Transition to a trained forecasting model once a minimum data window is available
- Incorporate holiday calendar and known scheduled events as contextual features

Phase 5 — App Integration & Demo (Final Weeks)

- Integrate the feature module into the existing Megaworld web app shell
- Build the trip-planning view (“will I likely find parking if I arrive at X time”)

- Prepare final presentation: architecture walkthrough, live demo, and a clear “what's next if this were expanded” roadmap

12. Risks & Mitigations

Risk	Impact	Mitigation
Limited camera/infra access during internship	Medium–High	Scope pilot to 1–2 rows; simulate additional data if needed for the ML demo
Insufficient historical data for ML model	High	Heuristic fallback (Phase 4); consider synthetic/simulated data clearly labeled as such for demo purposes
Plate-matching false positives	Medium	Confidence threshold + “unresolved” fallback state; never auto-resolve on a low-confidence or near-tie match
Data privacy/compliance concerns	Medium–High	Early legal/IT review; minimize plate data retention; keep pilot data sandboxed
Scope creep (reservations, pricing, navigation)	Medium	Explicit non-goals list (Section 4); treat as future roadmap, not internship deliverables

13. Suggested Technology Directions

The following are directional suggestions, not final commitments — to be validated against what Megaworld's existing stack and infrastructure already support.

- **Computer vision:** a lightweight object-detection model for occupancy (e.g., a compact detector suitable for near-real-time inference) paired with an off-the-shelf ANPR/OCR component for plate reads, rather than training plate recognition from scratch.
- **Prediction/ML:** classical time-series or gradient-boosted models (well-suited to tabular, seasonal data like occupancy-by-hour) as a first approach, before considering deep learning — simpler models are easier to explain, debug, and justify within an internship timeframe.
- **Backend/integration:** REST or event-based API layer so the occupancy/prediction service can be consumed by the existing web app without tightly coupling the two codebases.
- **Data storage:** a relational store for slot/zone metadata and transactional logs (occupancy transitions, match events), suitable for both operational queries and model training exports.

14. Future Enhancements (Post-Internship Roadmap)

- Expand camera coverage to full-building, multi-level rollout across a township
- Extend prediction model across multiple townships with cross-site demand correlation (e.g., Eastwood filling up increases likelihood of overflow demand at nearby sites)
- Push notifications: proactively alert users planning a trip if predicted availability drops below a threshold
- Map-based visualization showing availability across all Megaworld properties at a glance
- Optional slot reservation layer, if the business decides to move in that direction later

15. Summary

This project combines three technically distinct but complementary components — lightweight computer-vision occupancy sensing, resource-efficient approximate plate matching against the ticketing system, and ML-based demand forecasting — into a single, incrementally deployable feature for Megaworld's township app. Its phased structure is designed to produce a working, demonstrable milestone at every stage, so the internship deliverable remains meaningful even if later phases (particularly full ML forecasting) are constrained by data availability or timeline.